

# ENHANCING INDONESIAN DEEFAKE SPEECH LOCALIZATION WITH PATHOLOGICAL FEATURES

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## ABSTRACT

Deepfake speech detection and localization are critical for preserving audio integrity, particularly in the underexplored context of the Indonesian language. This study presents a hybrid Light Convolutional Neural Network (LCNN) and Bidirectional Long Short-Term Memory (BiLSTM) framework to detect and localize deepfake segments in Indonesian speech. A dedicated dataset was constructed consisting of (1) genuine Indonesian speech from open and closed sources, (2) fully spoofed speech generated using text-to-speech (TTS) and voice conversion (VC) models, and (3) partially spoofed speech created by inserting synthetic segments into genuine audio and vice versa. Input features include eight pathological features combined with Linear Frequency Cepstral Coefficients (LFCC). Although pathological features alone do not surpass LFCC, their combination significantly improves localization accuracy. Experimental results confirm the effectiveness of pathological features and the LCNN-BiLSTM architecture for Indonesian deepfake speech localization.

**Index Terms**— Deepfake speech localization, pathological features, LCNN-BiLSTM, Indonesian language

## 1. INTRODUCTION

Advances in Artificial Intelligence (AI) have accelerated rapidly, creating opportunities across multiple domains while also raising security risks. One of the most concerning misuses is deepfake, which enables highly realistic manipulation of audio, images, and video [1]. In particular, audio deepfakes have gained attention due to their accessibility on personal devices and their potential for malicious use [2]. Beyond fully synthesized speech, partially spoofed audio—where genuine and synthetic segments are combined—poses greater challenges as it is more realistic and harder to detect [3].

To address such threats, deepfake speech localization has emerged as a promising task. Unlike binary detection, localization identifies the exact temporal segments of audio

that are synthetic, which is especially important for partial spoofing [3]. Convolutional Neural Network (CNN)-based architectures are particularly well-suited for this task, as they can capture local spectral-temporal patterns in audio features. Among them, LCNN provides efficient feature extraction, while BiLSTM networks capture temporal dependencies. Their combination has demonstrated strong performance in deepfake detection and localization [4], [5].

While cepstral-based features such as LFCC have been widely adopted in deepfake detection [5], recent research suggests that pathological features can offer complementary cues by capturing fine-grained distortions in voice quality that are often introduced by synthesis [6]. These speech-pathological features are acoustic measures traditionally used in clinical voice analysis to distinguish normal from pathological voices [7] and have been applied to the diagnosis of many disorders including Parkinson's disease [8], head and neck cancers [9], and other organic pathologies [10]. They quantify subtle variations in pitch stability, amplitude consistency, and harmonic structure, making them particularly valuable for spoof detection and localization as synthesis systems cannot perfectly reproduce the natural imperfections of human speech.

However, applying these advances to Indonesian speech remains challenging because most available partial spoof datasets focus on English and Mandarin [11], [12], leaving a significant gap in resources for Indonesian partial deepfake research. To address this limitation, this study introduces an Indonesian deepfake speech localization system based on an LCNN-BiLSTM architecture that incorporates both LFCC and pathological features. The proposed approach aims to enhance the accuracy of detecting and localizing spoofed segments under partial spoofing conditions, while simultaneously contributing to Indonesian deepfake research through the creation of a dedicated dataset and comprehensive model evaluation.

## 2. RELATED WORKS

Previous research has explored the use of pathological features in deepfake speech detection. Systems that combine

features such as jitter, shimmer, and glottal-to-noise ratios with conventional spectral features, like LFCC, have demonstrated improved performance compared to using spectral features alone [6], [13]. Segment-level extraction and the integration of additional spectral representations, such as mel-spectrograms, have further enhanced detection accuracy and recall [6].

Beyond utterance-level detection, segment-level detection, or localization, has received increasing attention due to its relevance in partially spoofed audio. Localization systems often employ CNN-based architectures combined with recurrent networks to capture both spatial and temporal patterns in audio. These methods have shown that segment-level detection is more challenging than utterance-level detection but provide precise information on the positions of synthetic segments [5], [14].

Dataset availability remains a key challenge for segment-level deepfake detection. Existing datasets primarily focus on English and Mandarin and include segmental annotations for partially spoofed audio [5], [14]. For the Indonesian language, only a few resources have recently emerged, with the InaSpoof corpus providing a large-scale benchmark for antispoofing that contains diverse bona fide and spoofed speech but lacks partially spoofed audio and segmental annotations [15].

### 3. DATASET

To address the challenges of spoof localization in Indonesian speech, the dataset used in this study integrates bona fide, fully spoofed, and partially spoofed speech. Table 1 summarizes the variability and distribution of the dataset.

#### 3.1. Bona Fide

The bona fide subset was compiled from two Indonesian speech corpora: Common Voice [16] and Prosa.ai<sup>1</sup>. Common Voice is a widely used open-source dataset that contains diverse speaker accents and varying recording qualities, while audios from Prosa.ai is a closed-source dataset collected in a more controlled recording environment and made available through previous research.

Each utterance was paired with its transcript to enable accurate text alignment maintain accurate mapping between the audio and textual content. To ensure consistency and audio quality, the DNSMOS quality estimator was applied, and only samples with an overall quality (OVRL) score  $\geq 3.0$  were retained.

#### 3.2. Fully Spoofed

The fully spoofed subset was generated by synthesizing transcripts from the compiled bona fide audio recordings

**Table 1. Dataset Variability and Distribution**

| Dataset           | Dataset Source / Speech Synthesis | Utterance | Total Hours |
|-------------------|-----------------------------------|-----------|-------------|
| Bona fide         | Common Voice                      | 3,101     | 3.30        |
|                   | Prosa.ai                          | 1,288     | 2.04        |
| Fully spoofed     | Google TTS                        | 4,389     | 4.80        |
|                   | Facebook MMS                      | 4,389     | 4.87        |
|                   | VITS                              | 4,389     | 4.04        |
|                   | ElevenMultilingualV2              | 349       | 0.35        |
| Partially spoofed | Google TTS                        | 4,389     | 4.74        |
|                   | Facebook MMS                      | 4,389     | 4.46        |
|                   | VITS                              | 4,389     | 3.24        |
|                   | ElevenMultilingualV2              | 349       | 0.38        |

using four TTS systems: Google TTS, Facebook MMS, ElevenMultilingualV2, and a pre-trained VITS model<sup>2</sup>. The VITS model was obtained from Hugging Face and was pretrained on Indonesian speech data. ElevenMultilingualV2 was reserved exclusively as an unseen attack during evaluation to simulate cross-system generalization.

To further improve speaker similarity, voice conversion was applied using the pretrained FreeVC model from the official repository<sup>3</sup>, with bona fide utterances serving as reference voices. Similar to the bona fide subset, all synthesized audio was subsequently re-evaluated with DNSMOS to ensure perceptual quality consistency.

#### 3.3. Partially Spoofed

The partially spoofed subset was created to enable segment-level spoof localization. Two complementary scenarios were designed:

- Bona fide-to-Spoof (B2S), in which short genuine segments were inserted into spoofed audio
- Spoof-to-Bona fide (S2B), in which spoofed segments were embedded into genuine speech.

These two configurations simulate both real-within-fake and fake-within-real conditions, providing diverse and balanced examples that enhance the model's ability to accurately detect and localize synthetic boundaries.

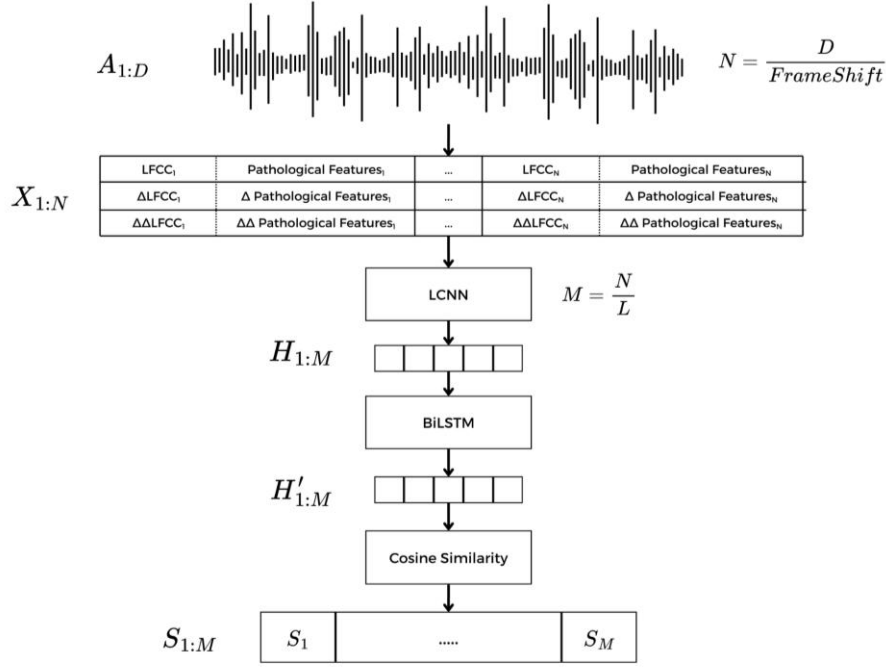
Candidate insertion points were determined by aligning transcripts with timestamps generated using Whisper ASR, followed by fuzzy word matching with RapidFuzz, applying a similarity threshold of 80%. The selected segments were volume-normalized and blended with crossfading and padding to reduce audible discontinuities. Metadata containing insertion timestamps and manipulation labels was recorded to provide segment-level ground truth.

Following the collection and construction process, the final dataset comprised 4,389 bona fide utterances, 13,516 fully spoofed utterances, and 13,513 partially spoofed utterances, resulting in a total of 31,418 audio samples. The

<sup>1</sup> Prosa.ai is an Indonesian AI company providing proprietary speech data and language technology resources

<sup>2</sup> <https://huggingface.co/jerichosiahaya/vits-tts-id>

<sup>3</sup> <https://github.com/OlaWod/FreeVC>



**Fig 1.** Segment-level spoof localization system. An input audio  $A_{1:D}$  is framed into  $N$  feature vectors resulting in  $X_{1:N}$  (LFCCs, pathological features, their delta and delta-delta terms). The LCNN downsamples the sequence by a factor  $L$  to obtain  $H_{1:M}$ , which is refined by a BiLSTM to  $H'_{1:M}$ . Cosine similarity with class vectors produces segment scores  $S_{1:M}$ , each covering a temporal window of  $R$ , enabling localization of spoofed versus bona fide regions

dataset and the source code used to generate the Indonesian deepfake speech localization corpus remain unavailable for public release due to licensing constraints and the inclusion of certain third-party models. A release of the bona fide portion along with metadata generation scripts is planned for future work, subject to data licensing compliance.

#### 4. FEATURE EXTRACTION

As this study focuses on exploring the role of pathological features for segment-level deepfake speech localization in the Indonesian language, which remains largely underexplored, we adopted a previously established feature framework combining the spectral feature LFCC with pathological features [13]. Thus, other features frequently used for spoof detection, such as MFCC and CQCC, were not used in order to maintain a clear focus on the contribution of pathological features and ensure comparability with prior work.

For this study, eight pathological features were extracted: Jitter (local, PPQ3, PPQ5), Shimmer (local, APQ3, APQ5, APQ11), and Glottal-to-Noise Excitation ratio (GNE). These features were computed using a segmental frame analysis approach, where the speech signal was divided into frames and each frame was analyzed independently. For each frame, all eight features were concatenated to form a single feature vector representing

that segment, capturing fine-grained temporal variations in the speech signal. To enrich the temporal dynamics, first-order and second-order derivatives of these pathological features were also computed and appended [6].

LFCC features were extracted using the same segment length and frame parameters as the pathological features to ensure temporal alignment. To further capture dynamic changes in spectral information, delta and delta-delta coefficients were derived and concatenated with the static LFCCs. Finally, both feature sets were combined at the segment level to produce a unified representation that incorporates complementary spectral and phonatory information.

#### 5. LOCALIZATION SYSTEM

To facilitate implementation, this study adopts an LCNN-BiLSTM architecture based on a model previously developed and validated in earlier research [5], which was an extension of the top single LCNN model in the ASVspoof 2019 LA task [17], [18]. This section explains how the model is trained to detect spoofed regions at the segment level.

The process begins with an input audio signal of duration  $D$  (in milliseconds), from which features are extracted as described in the previous section. Let the resulting feature sequence be denoted as  $X_{1:N} \equiv (X_1, \dots, X_N)$

where  $N$  is the number of frames extracted and  $X_n$  is the feature for the  $n$ -th frame. The number of frames  $N$  is determined by dividing the audio duration by the extraction frame shift used.

The feature sequence is first processed by the LCNN, which transforms the feature sequence  $X_{1:N}$  into a downsampled representation  $H_{1:M} \equiv (H_1, \dots, H_M)$ . The number of output frames  $M$  is determined by  $M = N/L$ , where  $L$  denotes the convolution stride of the LCNN and represents the downsampling factor, which following the design of the LCNN architecture, is 16. The resulting sequence is then passed through a BiLSTM layer, which captures temporal dependencies in both forward and backward directions, producing the enhanced representation  $H'_{1:M}$ .

Finally, the BiLSTM output representations are compared with class representation vectors using cosine similarity, producing segment-level scores  $S_{1:M}$ . Each score  $S_m$  reflects the similarity of the  $m$ -th segment to class  $k$ , with  $k=1$  indicating a spoofed segment and  $k=0$  indicating a bona fide segment. Based on these scores, the system generates segment-level predictions for each  $m$ -th segment, where each segment corresponds to a temporal window of  $R$  which is determined by multiplying the downsample factor  $L$  with the frame shift used during the feature extraction process. This design enables the system to localize spoofed and bona fide regions within an utterance, rather than producing only a global decision. The overall system workflow is illustrated in Fig. 1.

## 6. EXPERIMENTAL SETTINGS

Two experimental scenarios were designed to evaluate the effectiveness of the proposed approach. The first scenario involves training the model employing only pathological features, while the second combines pathological features with LFCC to examine the impact of feature fusion on model performance. In addition to these two scenarios, a baseline model employing only LFCC features was also developed to provide a comparison and assess the contribution of pathological features in the proposed framework.

For experimental evaluation, the dataset was divided into training, development, and testing sets following an 8:1:1 split ratio applied independently for each spoofing model and manipulation scenario. For the fully spoofed subset, separate splits were performed for Google TTS, Facebook MMS, and VITS outputs. Similarly, the partially spoofed subset was split by model and manipulation scenario to ensure balanced representation across systems and attack types. The ElevenMultilingualV2 system served as the sole exception and was reserved in its entirety as unseen test data to enable evaluation of cross-system generalization. A detailed summary of the dataset split, and distribution is provided in Table 2.

**Table 2. Dataset Split and Distribution**

| Dataset           | Dataset Source / Speech Synthesis | Utterance |     |      |
|-------------------|-----------------------------------|-----------|-----|------|
|                   |                                   | Train     | Dev | Test |
| Bona fide         | Common Voice                      | 2,480     | 310 | 311  |
|                   | Prosa.ai                          | 1,030     | 129 | 129  |
| Fully Spoofed     | Google TTS                        | 3,510     | 439 | 440  |
|                   | Facebook MMS                      | 3,510     | 439 | 440  |
|                   | VITS                              | 3,510     | 439 | 440  |
|                   | ElevenMultilingualV2              | 0         | 0   | 349  |
| Partially Spoofed | Google TTS                        | 3,510     | 439 | 440  |
|                   | Facebook MMS                      | 3,510     | 439 | 440  |
|                   | VITS                              | 3,510     | 439 | 440  |
|                   | ElevenMultilingualV2              | 0         | 0   | 349  |

For all feature extraction processes, including both LFCC and pathological features, a 20 ms analysis window and a 10 ms frame shift are applied. In the baseline LFCC configuration, a 512-point FFT and a 20-channel filter bank are used. In contrast, for the experimental setup that combines LFCC with pathological features, a 320-point FFT is applied. This adjustment ensures that the length of the LFCC features are consistent with the pathological features, thereby enabling effective concatenation.

The LCNN component adopted in this study is based on previous research that employed LCNN for segment-level detection tasks. Based on the used architecture, the LCNN will give a downsampling factor of  $L=16$ , allowing each prediction or embedding to be extracted every 160 ms [5]. This temporal resolution aligns with the segmentation of ground-truth labels, ensuring that both prediction outputs and labels are temporally consistent.

For training, the Adam optimizer ( $\beta_1=0.9$ ,  $\beta_2=0.999$ ,  $\epsilon=10^{-8}$ ) is employed with a batch size of 16. The initial learning rate is set to  $3 \times 10^{-4}$  and is reduced by half every 10 epochs to facilitate stable convergence [5]. The model is trained for a total of 50 epochs. All experiments are conducted in an NVIDIA Tesla V100 DGXS environment equipped with 32 GB of GPU memory, ensuring sufficient computational capacity for handling the feature extraction and model training processes.

Model performance is evaluated using a set of metrics, including accuracy, recall, F1-score, and Equal Error Rate (EER) [5], [6]. Evaluation is performed under two conditions: first, using the complete test dataset, and second, using only unseen test data. The unseen test data specifically consist of audio generated by the ElevenMultilingualV2 TTS system, allowing assessment of the model's generalization to novel synthetic speech.

## 7. EVALUATION RESULTS

As shown in Table 3, the model utilizing only pathological features achieved moderate performance, with a higher EER and lower accuracy and recall compared to the LFCC only

**Table 3.** Model Evaluation with All Test Data

| Features                   | Metrics (%) |              |              |              |
|----------------------------|-------------|--------------|--------------|--------------|
|                            | EER         | Accuracy     | Recall       | F1-score     |
| LFCC                       | 0.68        | 99.34        | 99.40        | 99.29        |
| Pathological               | 3.55        | 96.50        | 96.55        | 96.28        |
| <b>LFCC + Pathological</b> | <b>0.59</b> | <b>99.46</b> | <b>99.54</b> | <b>99.42</b> |

**Table 4.** Model Evaluation with Only Unseen Test Data

| Features                   | Metrics (%) |              |              |              |
|----------------------------|-------------|--------------|--------------|--------------|
|                            | EER         | Accuracy     | Recall       | F1-score     |
| LFCC                       | 1.06        | 98.97        | 98.99        | 98.82        |
| Pathological               | 5.22        | 93.75        | 92.78        | 93.02        |
| <b>LFCC + Pathological</b> | <b>0.90</b> | <b>99.12</b> | <b>99.15</b> | <b>99.00</b> |

baseline. This indicates that while pathological features capture fine-grained perturbations, they are not sufficient on their own to surpass conventional cepstral features. The LFCC only system delivered stronger results, achieving lower EER and higher F1-score, confirming the effectiveness of cepstral information for spoof detection and localization.

The integration of pathological features with LFCC, however, produced the best performance across all metrics. The hybrid system achieved the lowest EER and the highest F1-score, while also improving Accuracy and Recall. These gains suggest that combining spectral and pathological cues reduces both false positives and false negatives, making the model more reliable in detecting partially spoofed audio.

To further test robustness, the models were evaluated on the unseen ElevenMultilingualV2 subset. As shown in Table 4, the pathological-only system again fell short of the LFCC baseline, confirming its limitations when applied in isolation. In contrast, the LFCC and pathological hybrid continued to achieve the highest performance, maintaining improvements over both individual models even under domain shift. This result demonstrates the generalization capability of the proposed feature integration.

Overall, the evaluation demonstrates that while pathological features alone provide limited benefits, their combination with LFCC substantially enhances model performance. The hybrid system achieves superior results on both the full test set and the unseen dataset, confirming the potential of pathological features in challenging scenarios involving partially spoofed audio, and highlighting the value

of integrating complementary feature types for a more effective and robust deepfake speech localization.

## 8. CONCLUSION

This work developed an Indonesian deepfake speech localization system based on an LCNN-BiLSTM architecture with both LFCC and pathological features. The results show that while pathological features alone were less effective than LFCC, their integration significantly improved detection and localization performance, including under unseen attack scenarios. These findings confirm that pathological cues serve as valuable complements to cepstral features for enhancing effectiveness and robustness.

Future work should expand the dataset by incorporating a wider range of synthesis methods, including using more text-to-speech and voice conversion models, to capture the evolving landscape of deepfake generation. In addition, collecting speech from speakers representing diverse regional dialects, accents, and speaking styles would help the dataset better reflect real-world variability. These expansions would provide richer training data, expose the model to a greater variety of synthetic artifacts, and ultimately strengthen its ability to generalize across unseen attack types and real-world deployment scenarios.

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## REFERENCES

- [1] F. F.-H. Nah, R. Zheng, J. Cai, K. Siau, and L. Chen, “Generative AI and ChatGPT: Applications, challenges, and AI-human collaboration,” *Journal of information technology case and application research*, vol. 25, no. 3, pp. 277–304, Jul. 2023.
- [2] Z. Almutairi and H. Elgibreen, “A Review of Modern Audio Deepfake Detection Methods: Challenges and Future Directions,” *Algorithms*, vol. 15, no. 5, p. 155, May 2022.
- [3] L. Zhang et al., “Spoof Diarization: ‘What Spoofed When’ in Partially Spoofed Audio,” *arXiv (Cornell University)*, Sep. 2024.
- [4] A. Mittal and M. Dua, “Automatic speaker verification systems and spoof detection techniques: review and analysis,” *International Journal of Speech Technology*, vol. 25, no. 1, pp. 105–134, Aug. 2021.
- [5] L. Zhang, X. Wang, E. Cooper, J. Yamagishi, J. Patino, and N. Evans, “An Initial Investigation for Detecting Partially Spoofed Audio,” *Interspeech 2022*, pp. 4264–4268, Aug. 2021.

- [6] Anuwat Chaiwongyen, Suradej Duangpummet, Jessada Karnjana, Waree Kongprawechnon, and Masashi Unoki, "Potential of Speech-Pathological Features for Deepfake Speech Detection," *IEEE Access*, vol. 12, pp. 121958–121970, Jan. 2024.
- [7] A. Sasou, "Automatic identification of pathological voice quality based on the GRBAS categorization," in *Proc. Asia-Pacific Signal Inf. Process. Assoc. Annu. Summit Conf. (APSIPA ASC)*, Dec. 2017, pp. 1243–1247.
- [8] D. Meghraoui, B. Boudraa, T. Merazi-Meksen, and P. G. Vilda, "A novel pre-processing technique in pathologic voice detection: Application to Parkinson's disease phonation," *Biomed. Signal Process. Control*, vol. 68, Jul. 2021, Art. no. 102604.
- [9] R. Islam, M. Tarique, and E. Abdel-Raheem, "A survey on signal processing based pathological voice detection techniques," *IEEE Access*, vol. 8, pp. 66749–66776, 2020.
- [10] S. R. Kadiri and P. Alku, "Analysis and detection of pathological voice using glottal source features," *IEEE J. Sel. Topics Signal Process.*, vol. 14, no. 2, pp. 367–379, Feb. 2020.
- [11] N. M. Müller et al., "MLAAD: The Multi-Language Audio Anti-Spoofing Dataset," *2022 International Joint Conference on Neural Networks (IJCNN)*, vol. 33, pp. 1–7, Jun. 2024.
- [12] W. Mustikarini, R. Hidayat, and A. Bejo, "Real-Time Indonesian Language Speech Recognition with MFCC Algorithms and Python- Based SVM," *IJITEE (International Journal of Information Technology and Electrical Engineering)*, vol. 3, no. 2, p. 55, Oct. 2019.
- [13] Anuwat Chaiwongyen, Suradej Duangpummet, Jessada Karnjana, Waree Kongprawechnon, and Masashi Unoki, "Deepfake-speech Detection with Pathological Features and Multilayer Perceptron Neural Network," *2022 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*, pp. 2182–2188, Oct. 2023.
- [14] J. Yi et al., "Half-Truth: A Partially Fake Audio Detection Dataset," *arXiv.org*, Dec. 15, 2023.
- [15] C. O. Mawalim, S. A. Arief, and D. P. Lestari, "InaSAS: Benchmarking Indonesian Speech Antispoofing Systems," *APSIPA Transactions on Signal and Information Processing*, vol. 14, e203, 2025.
- [16] R. Ardila, M. Branson, K. Davis, M. Henretty, M. Kohler, J. Meyer, R. Morais, L. Saunders, F. Tyers, and G. Weber, "Common Voice: A Massively-Multilingual Speech Corpus," in *Proc. 12th Conf. Lang. Resources and Evaluation (LREC 2020)*, Marseille, France, May 2020, pp. 4218–4222.
- [17] A. Nautsch et al., "ASVspooF 2019: Spoofing Countermeasures for the Detection of Synthesized, Converted and Replayed Speech," *IEEE Transactions on Biometrics, Behavior, and Identity Science*, vol. 3, no. 2, pp. 252–265, Apr. 2021.
- [18] G. Lavrentyeva, S. Novoselov, A. Tseren, M. Volkova, A. Gorlanov, and A. Kozlov, "STC Antispoofing Systems for the ASVspooF2019 Challenge," *Interspeech 2019*, Sep. 2019.